

AI Usage Disclosure Form

1. Team Details

Team Name: Ultras

Project / Product Name: Prelude, Streaming Live RAG

Organization / Institution (if any): Thapar Institute of Engineering and Technology

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any AI / Gen AI tools or services in developing or creating project artifacts? Yes / No

YES

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Partial - Product positioning, feature set, gate strategy and architecture decided by team; AI used as a sounding board on trade-offs and to survey current RAG techniques

Code generation or assistance: Yes, Implementation assistance, boilerplate, test scaffolding

UI / UX design: Partial - Dashboard requirements, panel priorities and demo flow set by team; AI drafted the design brief (layout, visual tokens) and produced React/TypeScript components

Content creation: Yes, Documentation layout, README formatting, first drafts of documentation; demo corpus generated with AI and hand-checked by team; all reported numbers come from the committed evaluation harness

Data analysis: No, Evaluation metrics and methodology designed by team

Testing / debugging: Partial - AI suggested fixes; team found and resolved the real bugs through end-to-end testing (citation-id collision between document versions, hash chain breaking on restart, entity fallback on lowercase transcripts)

Other: Code refactoring suggestions, Python and TypeScript type definitions, prompt templates co-written and versioned in prompts/

4. Feature Origin Classification (Please add/delete based on the number of the feature)

4.1 Feature 1

4.1.1. Feature Name: **Retrieval Controller (Early-Retrieval Trigger)**

4.1.2. Self-Generated/AI-Generated/Both: **Both**

4.1.3. Description:

- **Team Contribution:** Controller concept, feature set (entity anchors, embedding drift, clause boundary, content-token minimum), Wait/Retrieve/No-Retrieval policy with reason codes, threshold tuning on labelled streams
- **AI Tools Used:** Claude Code
- **AI Assistance:** Python implementation of the feature extractor and policy module, FastAPI wiring
- **Modifications:** Team fixed the entity-extraction fallback that never fired on lowercase spoken transcripts, found through live mic testing

4.2 Feature 2

4.2.1. Feature Name: **Multi-Intent Decomposer**

4.2.2. Self-Generated/AI-Generated/Both: **Both**

4.2.3. Description:

- **Team Contribution:** Decomposition contract (strict JSON, cap of 4, near-duplicate merge), session-entity carry-over, prompt design
- **AI Tools Used:** Claude Code
- **AI Assistance:** JSON parsing and validation code, offline fallback provider
- **Modifications:** Team defined the rule that drops filler fragments as sub-queries

4.3 Feature 3

4.3.1. Feature Name: **Hybrid Retrieval Engine (BM25 + Dense + RRF + Rerank)**

4.3.2. Self-Generated/AI-Generated/Both: **Both**

4.3.3. Description:

- **Team Contribution:** Pipeline design, model selection (bge-small, MiniLM cross-encoder), fusion strategy, hybrid-vs-dense ablation methodology
- **AI Tools Used:** Claude Code
- **AI Assistance:** BM25 index code, Qdrant integration, rerank wiring
- **Modifications:** Team audited the pipeline against the design and had the missing rerank stage wired in

4.4 Feature 4

4.4.1. Feature Name: **Corpus Importer & Section-Aware Chunking**

4.4.2. Self-Generated/AI-Generated/Both: **Both**

4.4.3. Description:

- **Team Contribution:** [Doc_ID §Section] citation scheme, chunking strategy, version and effective-date metadata, demo corpus hand-checked and corrected
- **AI Tools Used:** Claude Code, Claude
- **AI Assistance:** Markdown chunker, ingestion code, first drafts of the synthetic demo corpus
- **Modifications:** Team resolved a citation-id collision between document versions that let a superseded document corrupt retrieval ranking

4.5 Feature 5

4.5.1. Feature Name: **Claim-Based Synthesis**

4.5.2. Self-Generated/AI-Generated/Both: **Both**

4.5.3. Description:

- **Team Contribution:** Claim schema (text, citation, verbatim quote), prose-rendered-from-claims rule, synthesis prompt
- **AI Tools Used:** Claude Code
- **AI Assistance:** Prompt wiring, streamed response parsing
- **Modifications:** Team defined the minimum-overlap rule after catching a false positive on an off-corpus question

4.6 Feature 6

4.6.1. Feature Name: **Grounding Verifier**

4.6.2. Self-Generated/AI-Generated/Both: **Both**

4.6.3. Description:

- **Team Contribution:** Subtractive design (ID allow-list, verbatim quote match, NLI entailment), rule that the verifier can only reject, never edit or invent
- **AI Tools Used:** Claude Code
- **AI Assistance:** Verifier pipeline implementation, NLI model integration
- **Modifications:** Team specified a reason code for every rejection and the behaviour on the injected adversarial chunk

4.7 Feature 7

4.7.1. Feature Name: **Session Claim Graph & Delta Refinement**

4.7.2. Self-Generated/AI-Generated/Both: **Both**

4.7.3. Description:

- **Team Contribution:** Refine-not-restart concept, claim-graph model, affected-claim detection, v1 to v2 diff contract, 30-minute TTL policy
- **AI Tools Used:** Claude Code
- **AI Assistance:** Claim graph implementation, Redis persistence
- **Modifications:** Team required the TTL to be proven by a test and mirrored in the UI

4.8 Feature 8

4.8.1. Feature Name: **Presentation-Turn Suppression & Session Semantic Cache**

4.8.2. Self-Generated/AI-Generated/Both: **Both**

4.8.3. Description:

- **Team Contribution:** Zero-retrieval re-render concept, cache scope (session only) and threshold (cosine ≥ 0.9)
- **AI Tools Used:** Claude Code
- **AI Assistance:** Phrase detector, cache module
- **Modifications:** Team tuned the presentation-turn limit after live mic testing

4.9 Feature 9

4.9.1. Feature Name: **Hash-Chained Telemetry & Cost Meter**

4.9.2. Self-Generated/AI-Generated/Both: **Both**

4.9.3. Description:

- **Team Contribution:** Event schema, trace-id flow, hash-chain design, cost accounting rules
- **AI Tools Used:** Claude Code
- **AI Assistance:** Hash chain code, JSONL writer, verify endpoint
- **Modifications:** Team fixed the chain breaking on service restart

4.10 Feature 10

4.10.1. Feature Name: **Security Layer**

4.10.2. Self-Generated/AI-Generated/Both: **Both**

4.10.3. Description:

- **Team Contribution:** Threat model T1-T10, HMAC session tokens, PII redaction policy, no-hardcoded-eval rule
- **AI Tools Used:** Claude Code
- **AI Assistance:** nginx config, token code, CI jobs
- **Modifications:** Team replaced a fixed grep with a check derived from the evaluation streams

4.11 Feature 11

4.11.1. Feature Name: **Live Dashboard (Transcript, Headroom Ruler, Claim Diff, Evidence Drawer, Evidence Graph)**

4.11.2. Self-Generated/AI-Generated/Both: **Both**

4.11.3. Description:

- **Team Contribution:** Dashboard requirements (which controller decisions and demo moments must be visible on screen), panel priorities, review and revision of the design brief against the positioning doc
- **AI Tools Used:** Claude Code, Claude

- **AI Assistance:** Design brief (layout, visual tokens, keyboard model, demo-moment list), React component boilerplate, SVG rendering, Vitest setup
- **Modifications:** Team decided headroom in chunks instead of milliseconds, clause-sized mic chunks, and a hand-laid evidence graph instead of a library

4.12 Feature 12

4.12.1. Feature Name: **Evaluation Framework (G1-G6 Scorecard)**

4.12.2. Self-Generated/AI-Generated/Both: **Both**

4.12.3. Description:

- **Team Contribution:** Complete methodology, stabilisation-headroom metric, stream categories and labelling, false-positive analysis, ablation design
- **AI Tools Used:** Claude Code
- **AI Assistance:** Harness scripts, scorecard rendering
- **Modifications:** Team corrected mislabelled safe points and interpreted results

4.13 Feature 13

4.13.1. Feature Name: **Record/Replay Modes & Microservices Packaging**

4.13.2. Self-Generated/AI-Generated/Both: **Both**

4.13.3. Description:

- **Team Contribution:** live/record/replay/offline mode design, service boundaries, healthcheck design, guided tour script, privacy stance
- **AI Tools Used:** Claude Code
- **AI Assistance:** Replay provider, Dockerfiles, compose file, CI workflows
- **Modifications:** Team optimized the default image to build without model downloads and set the demo tour flow

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Saumil Makkar

Role: AI Developer

Date: 30 September 2026